

Title

Control-impact architecture of the human structural connectome: degree dominance, a small near-universal residual, and a cross-scale architectural comparison with the fly connectome

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Ethics: Secondary analysis of publicly distributed, already-published data (AOMIC-ID1000, Zenodo 19796783, CC-BY-4.0; FAFB v783, CC BY-NC 4.0). No new human data were collected; per the author’s determination of 2026-09-26, no IRB approval or exemption is required for this secondary use (see SUBMISSION_PACKAGE/ethics_statement_if_applicable.md).

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Abstract

Background. Network communication in brain connectomes is dominated by node degree, but whether any degree-independent component of node-level control impact exists — and whether it is anatomically structured — remains unresolved, because most node-removal analyses are group-average, lack matched controls, and lack null-model arbitration. **Gap.** No prior study, to our knowledge, has combined exact per-node removal-based control impact with per-subject population-scale statistics, degree/strength matching, degree-preserving null ensembles, and a pre-registered cross-scale comparison. **Data and methods.** We computed the exact Control Impact Score, $CIS(i) = (E(G) - E(G-i))/E(G)$, for all 456 nodes in each of 801 QC-pass subjects from the AOMIC-ID1000 structural-connectome resource ($N = 900$ acquired; 456-node 4S456 parcellation; SIFT streamline counts; 15% proportional threshold), with degree-matched controls ($\pm 10\%$), 100 degree-preserving (Maslov–Sneppen) nulls per graph in two batteries (100×100 full-CIS; 200×100 top-50), BH-FDR node-level inference, system-enrichment tests with two control levels, and a 9-configuration robustness ladder. **Main human result.** Top-50 CIS concentration exceeded its degree-preserving null in 200/200 subjects (median $z = 16.36$; two-sided sign $p = 1.24 \times 10^{-60}$). CIS was strongly degree-related (median $\rho = 0.943$). A small degree-independent residual was present in 778/801 subjects (sign $p = 2.6 \times 10^{-197}$; Cliff’s $\delta = 0.100$; median Wilcoxon $p = 0.167$) and 43/456 nodes survived FDR ($q < .05$; max Stouffer $z = 23.3$). **Cross-scale result.** Compared with the frozen fly (FAFB v783) study — Cliff’s $\delta = 0.0979$, itself not surviving its degree-preserving null ensemble — the human architecture is concordant in effect size and degree dominance. **Anatomical divergence.** The residual’s anatomical identity differs: visual systems hold 2% of the human top 50 versus 80% visual-centrifugal in the fly; the single nominal human system signal (subcortical/cerebellar, $z_B = 2.08$, $p = .031$, $1.25\times$) is absent at $K = 25$ and degree-anchored, and the pre-registered system-enrichment gate (R2b) was **negative**. **Interpretation.** Control-impact architecture — a dominant degree component plus a small, near-universal degree-independent residual — shows cross-scale architectural similarity while the residual’s anatomical identity diverges. No new biological mechanism, no homology, and no system-specific claim is made.

1. Introduction

- **Established knowledge.** Node-removal/efficiency analysis in human structural connectomes is classic (Alstott et al. 2009; Crossley et al. 2014). Degree is the dominant correlate of node importance ($\rho \approx 0.9$ here).
- **Unresolved problem.** Whether a degree-independent component of control impact exists at the single-subject level, is reproducible across a population, and is anatomically structured — prior group-average designs cannot answer this.
- **Degree confounding.** Raw comparisons of node importance are degree-confounded; matched controls and degree-preserving nulls are both required (the fly study’s decisive lesson, re-confirmed here).
- **Literature gap.** The four-way combination (per-node CIS + per-subject + population scale + degree-preserving null arbitration) is unoccupied (NOVELTY_MATRIX.md).
- **Purpose and hypotheses.** H1: concentration exceeds degree-preserving null expectations. H2: a degree-independent residual remains after degree/strength adjustment. H3: the residual is robustly system-specific. H4: comparable architecture appears across fly and human scales. Pre-registered in PROTOCOL_FREEZE.md before any computation [refs 1, 3, 6, 8; dataset: 7].

2. Methods

Condensed; full frozen detail in ANALYSIS_FREEZE.md and supplementary.

- **Dataset and QC.** AOMIC-ID1000 derivatives (Zenodo 19796783, CC-BY-4.0) [7]; 900 subjects $\times$ 7 atlases extracted and verified (25,200 matrices; 0 NaN/Inf/asymmetry); QC cohort 801/900 (frozen CIS-blind; flags retained, never deleted).
- **Primary configuration.** 4S456Parcels; sift_radius2_count_connectivity; 15% proportional threshold ($k = 15,561$ edges).
- **CIS.** Exact per-node CIS = $(E(G) - E(G-i))/E(G)$; 456 exact leave-one-out recomputations per subject; reference implementation validated to $6.1e-16$; fast estimator (null graphs only) to $4e-16$ under a mandatory $1e-9$ gate.
- **Matching.** $\pm 10\%$ total degree, 1:1 greedy, nearest-50 fallback (flagged).
- **Null batteries.** Maslov–Sneppen undirected rewiring [6] preserving the exact per-node degree sequence (verified per null, 10,000/10,000 exact, 0 self-loops); seeds 100+i; subject samples nested, seed 20260922.
- **Statistics.** Per-node empirical p (add-one) $\rightarrow$ Stouffer $\rightarrow$ BH-FDR ($q=.05$); battery $B z + \text{sign test}$; enrichment controls A/B (10,000 perms); rank stability (100 half-splits); effect sizes Cliff’s δ .
- **Cross-scale discipline.** Normalized comparators only (DIRECT/ NORMALIZED/QUALITATIVE tags); no homology claims (CP03/CP05).

3. Results

(order per submission checklist)

1. **Data/QC.** 900 acquired, 801 pass; E0 median 0.5468 [0.5441–0.5497]; mean degree 68.2; 15,561 edges.
2. **CIS distribution.** Max population-mean CIS 0.00462 (node 414; e03_summary.json cis_mean_pop_max = 0.0046246); top-50 rate 1.00 for node 400; small magnitudes at 15% cost as expected for a redundant 68-degree-mean graph. *[Value corrected 2026-09-26 from “0.00463” — a rounding slip; the frozen artifact gives 0.0046246 $\rightarrow$ 0.00462 at 3 s.f.]*
3. **Degree relationship.** Median $\rho(\text{CIS}, \text{degree}) = 0.943$; CIS vs strength $\rho = 0.486$.
4. **Degree-preserving null test (H1).** 200/200 subjects above null; median $z = 16.36$; sign $p = 1.24e-60$; 100% of subjects at the 1/101 empirical- p floor (inference via $z + \text{sign test}$).
5. **Residual analysis (H2).** 778/801 subjects positive (sign $p = 2.6e-197$); $\delta = 0.100$ [0.078–0.129]; 36,846 matched pairs; 58.8% of comparisons beat controls; Wilcoxon median $p = 0.167$ ($125/801 < .05$) — small but directionally near-universal.
6. **FDR node characterization (R2a).** 43/456 nodes $q < .05$; max $z = 23.30$ (node 400); cluster spans nodes 400–451.

7. **System/enrichment (H3/R2b).** Pre-registered gate NEGATIVE: Vis obs 1 (2%), $z_B = -0.38$; only Subcortical/Cerebellar nominal (obs 26, z_A 9.06, z_B 2.08, p .031, 1.25 $\times$), absent at $K = 25$ (z_B 0.81) — degree-anchored.
8. **Robustness.** 9 configurations reproduce the primary architecture: max like-for-like top-50-mean deviation 0.00262 (atlas_AAL116; descriptive comparison — no numeric tolerance was pre-registered for E06); rank stability median Spearman 0.996.
9. **Cross-species comparison (H4).** δ 0.100 (human) vs 0.0979 (fly); both degree-dominated; human visual share 2% vs fly 80% visual-centrifugal — architecture replicates; anatomy does not.

4. Discussion

1. **Principal findings.** Degree-dominated control impact with a small, near-universal degree-independent residual; topological concentration exceeds degree-preserving nulls in every tested subject.
2. **Degree-dominated architecture.** $\rho \approx 0.94$ replicates the known degree–importance coupling at population scale, now with per-subject resolution.
3. **Degree-independent residual.** Small ($\delta = 0.100$), directionally near-universal (778/801), and measurable above the CIS noise floor only through matched controls plus null arbitration.
4. **Population-level concentration.** The 200/200 result with $z \approx 16$ shows concentration is a stable property of individual connectomes, not a group-average artifact.
5. **Cross-scale comparison.** Same effect-size statistic, same sign, same degree-anchored interpretation across a roughly 10^3 -fold node-count difference (139,255 reconstructed neurons vs 456 parcels).
6. **Architecture replicates but anatomy diverges.** Fly: visual-centrifugal residual (80% of top 50). Human: not visual (2%); nominal subcortical/ cerebellar signal is degree-anchored and K-ladder-fragile. **R2b was negative**; the present data do not establish that the mechanism selecting residual chokepoints is conserved (nor that it is not).
7. **Relationship to previous literature.** Alstott/Crossley [1, 3]: classic removal analysis, group-average, no matched controls or nulls. Kudriavtsev 2026 [5]: closest neighbour (lesions + nulls) but matched-mass nulls and ageing framing — PARTIAL OVERLAP (10_LITERATURE/). NCT literature [2, 4]: different metric. Yueh-Hsin 2024 [11]: resection simulations, no matching/nulls.
8. **Biological interpretation.** None beyond architecture: no new mechanism is claimed; no homology is claimed; the subcortical localization is degree-anchored; the fly–human δ similarity is interesting but partly fortuitous (different units, graphs, designs; fly δ not null-surviving).
9. **Methodological implications.** Individual methods are established; the contribution is the bounded combination and its arbitration. Matched-pair significance without null arbitration can mislead — both layers of control are necessary (fly E10B lesson, replicated).
10. **Limitations.** Undirected graphs (directed variant impossible in these derivatives); no homology; single pipeline/acquisition; residual near the CIS noise floor (hence null arbitration); battery-B per-subject p-values at the 1/101 null-resolution floor (inference via z + sign test); subcortical residual not robust across the K ladder; Kudriavtsev 2026 full text verified via proxy extraction (direct access blocked); ENVIRONMENT_MANIFEST provenance gap disclosed.
11. **Future work.** Directed human connectomes; multi-shell/higher-quality tractography; replication in BANC/larval connectomes and independent human cohorts; test of the two-component architectural hypothesis beyond two species.

5. Contribution and Novelty

A. Established methods. Node-removal/efficiency analysis of structural connectomes [1, 3; degree-ordered removal on the fly connectome: this project’s frozen predecessor], hub/rich-club analysis [8], degree-preserving null models [6], network controllability [2, 4], and cross-species connectome comparison [9; larval–adult *Drosophila* null + attack analyses: 10] are each established.

B. Methodological integration. To our knowledge, no prior study has combined per-node removal-based network-impact analysis, population-scale individual human structural connectomes, explicit degree-controlled

residualization, degree-preserving null-model arbitration, FDR-controlled residual inference, and a pre-specified cross-species comparison with the *Drosophila* connectome in a single analytical framework. The novelty therefore lies primarily in the integration of these analyses and the resulting cross-scale empirical comparison, rather than in any individual analytical component.

C. New empirical findings. (i) Top-50 CIS concentration exceeds degree-preserving null expectations in 200/200 tested subjects; (ii) a small degree-independent residual is detectable in 778/801 subjects ($\delta = 0.100$) with 43/456 nodes surviving FDR; (iii) the residual is not robustly system-specific (pre-registered gate negative).

D. Cross-species interpretation. A qualitatively similar two-component organization — a dominant connectivity-dependent component plus a smaller degree-independent residual — appears in both the human and *Drosophila* analyses despite roughly three orders of magnitude difference in analyzed node count and non-homologous anatomy. These results motivate the hypothesis that two-component control-impact organization recurs across nervous systems of very different scale; they do not establish anatomical homology, a conserved biological mechanism, or a universal law.

E. Limitations of the novelty claim. The claim is bounded by the documented literature audit (10_LITERATURE/NOVELTY_MATRIX.md; E00/E17/ E17a plus the 2026-09-26 updates). The closest neighbour (Kudriavtsev et al. 2026) has been full-text verified via proxy extraction as PARTIAL OVERLAP, with none of the four decisive components present (no per-node removal ranking, no degree-matched controls, no degree-preserving nulls, no cross-scale bridge). The audit cannot exclude occupation of the combination in sources not surfaced by the queries used. Full audit: 10_REPORT/NOVELTY_AUDIT.md.

6. Negative and Null Findings

The following negative results are first-class outcomes of this study and are preserved as such:

1. **Human system-specific gate (R2b) NEGATIVE.** No pre-specified sensory/visual system was enriched in the residual (Vis $z_B = -0.38$ at $K = 50$). The single nominal signal (Subcortical/Cerebellar: $z_B = 2.08$, $p = .031$, $1.25\times$) is absent at $K = 25$ ($z_B = 0.81$) and degree-anchored. No robust canonical-system-specific explanation of the residual was established (H3 not supported).
2. **Fly GABA/inhibitory-specific hypothesis REJECTED.** The raw matched-pair GABA–ACh effect ($\delta = 0.111$, $p = 0.0011$) did not survive the degree-preserving null ensemble (observed $\delta = 0.098$ at matched panel size vs null 0.071 ± 0.022 ; empirical $p = 0.109$; median-difference $p = 0.782$; $z = 1.21$). This methodological lesson — matched-pair significance can dissolve under degree-preserving nulls — motivated the two-layer control design used here.
3. **Fly residual effect size not null-surviving.** The fly residual $\delta = 0.0979$ itself did not pass the same null criterion ($p = 0.109$); the cross-scale δ agreement is therefore reported as interesting but partly fortuitous (different units of analysis, graphs, and designs).
4. **Per-subject empirical p-values (battery B) do not resolve the tail.** All sit at the 1/101 resolution floor; per-subject inference rests on the z -vs-null statistic and the population sign test.
5. **Subcortical residual is not robust across the pre-registered K ladder** and its FDR cluster overlaps the same node block as the nominal enrichment — R2a and R2b are not independent confirmations.
6. **No causal, homology, mechanism, or universal-law claim is made** at any point in this manuscript; the cross-scale similarity is an empirical architectural observation and a hypothesis-generating finding.

7. Conclusion

Across 801 individual human structural connectomes and the *Drosophila* connectome, structural network impact decomposes into a dominant connectivity-dependent component and a smaller degree-independent residual. Top-50 impact concentration exceeds degree-preserving null expectations in every tested human subject; the residual is directionally near-universal (778/801; $\delta = 0.100$) with 43/456 nodes surviving FDR; and the residual is not robustly system-specific. The qualitatively similar two-component organization at both scales — despite roughly

three orders of magnitude difference in analyzed node count and non-homologous anatomy — supports a bounded cross-scale architectural observation: architecture replicates; anatomy does not. The finding motivates the hypothesis that two-component control-impact organization may recur across nervous-system scales, and defines the null-arbitration methodology needed to test that hypothesis on future connectomes.

8. Data and code availability

Data: AOMIC-ID1000 (Zenodo 19796783, CC-BY-4.0). Fly reference: frozen release v1.0.0 artifacts (read-only). All human-side code, manifests, validation reports, and append-only logs in github.com/harsha-varadhan-2006/humanbrain_cross_species_cis; every number traces to an artifact (FINAL_REPRODUCIBILITY_AUDIT.md).

References

Inline citations use [n] keyed to this list; literature provenance and verdicts per study are frozen in 10_LITERATURE/NOVELTY_MATRIX.md and 10_LITERATURE/KUDRIAVTSEV_2026_FULLTEXT_REVIEW.md.

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